Class_5_Database_w_ggplot2

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Background

There are many *graphics systems* in R for making plots and figures. These include **"base R graphics"** like the `plot()` and add-on packages like **ggplot2**.

Let's compare how we make a simple graph with these two systems:

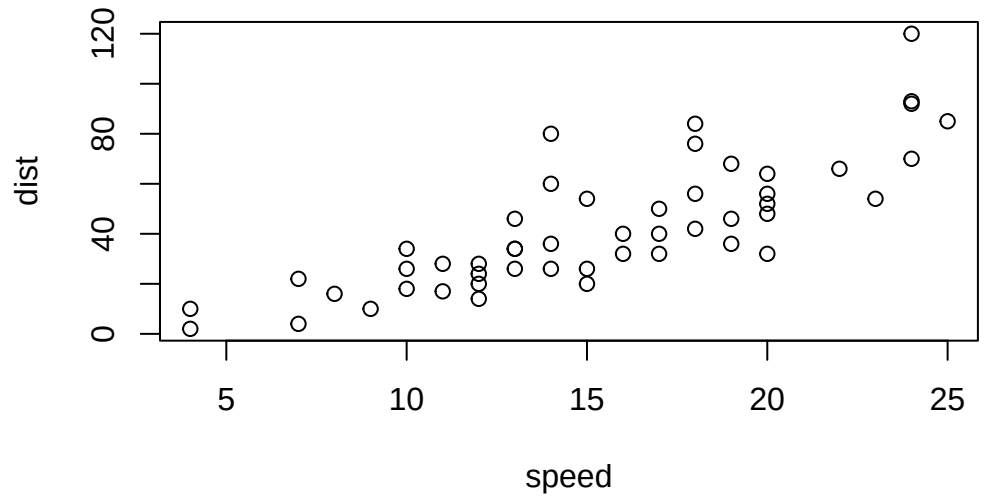
We can use the in-built `cars` dataset"

```
head(cars)
```

```
  speed dist
1     4    2
2     4   10
3     7    4
4     7   22
5     8   16
6     9   10
```

A base R plot:

```
plot(cars)
```



Now for **ggplot2**.

Before I can use ggplot2, I need to install it on my computer. To do this, we can use the function `install.packages("ggplot2")`

> **N.B.** We never run an `install.packages()` in our quarto doc (we run it in our R console) as it would re-install the package every time I render my document.

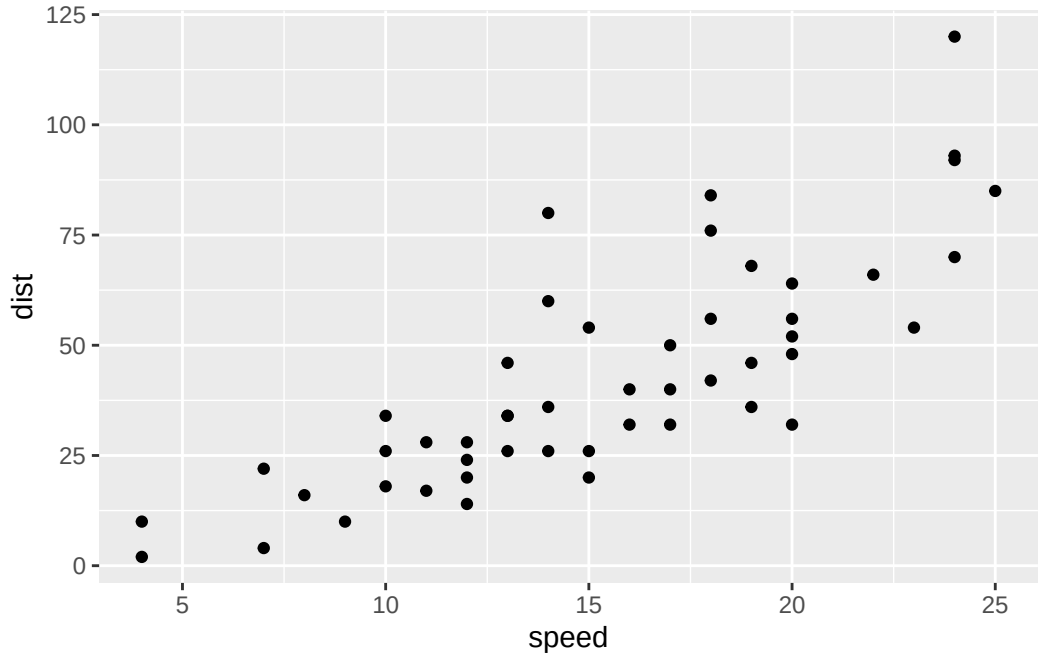
Once installed, we still need to load up the package into our R session. We do this with a `library()` call

```
library(ggplot2)
ggplot(cars)
```

Every ggplot needs at least 3 inputs:

- the **data** (a data.frame of the studd we want to plot)
- the **aesthetics** (how the data maps to the plot)
- the **geom** layer (how you want the plot drawn, points, lines, etc.)

```
ggplot(cars) +
  aes(x=speed, y=dist) +
  geom_point()
```

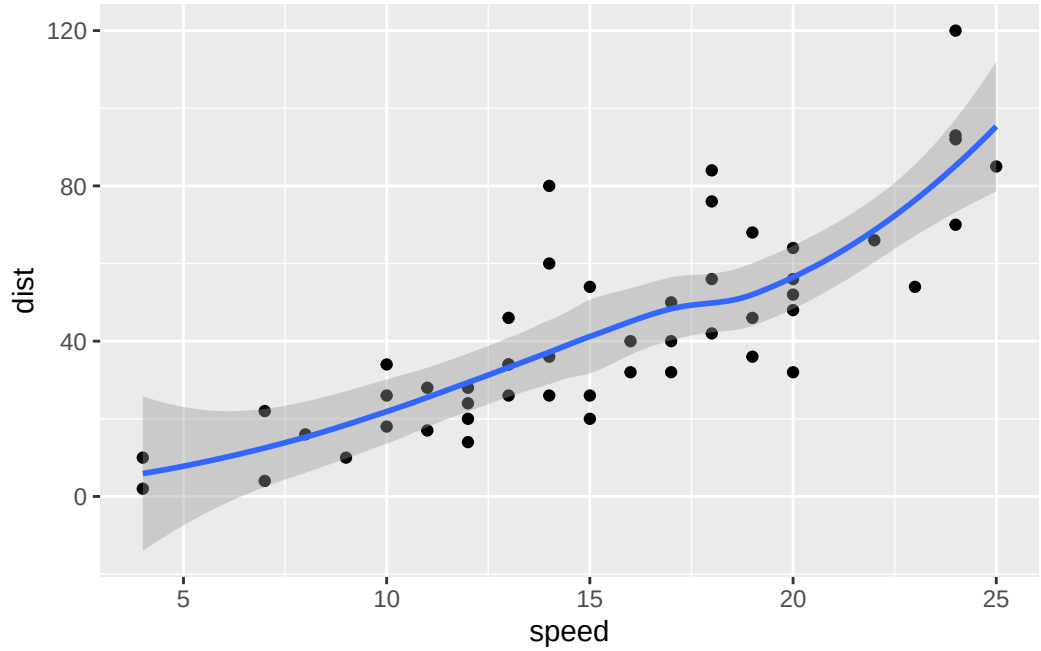


Let's add some features (more layers)

Let's add a trend line that shows the relationship between `speed` and `distance`:

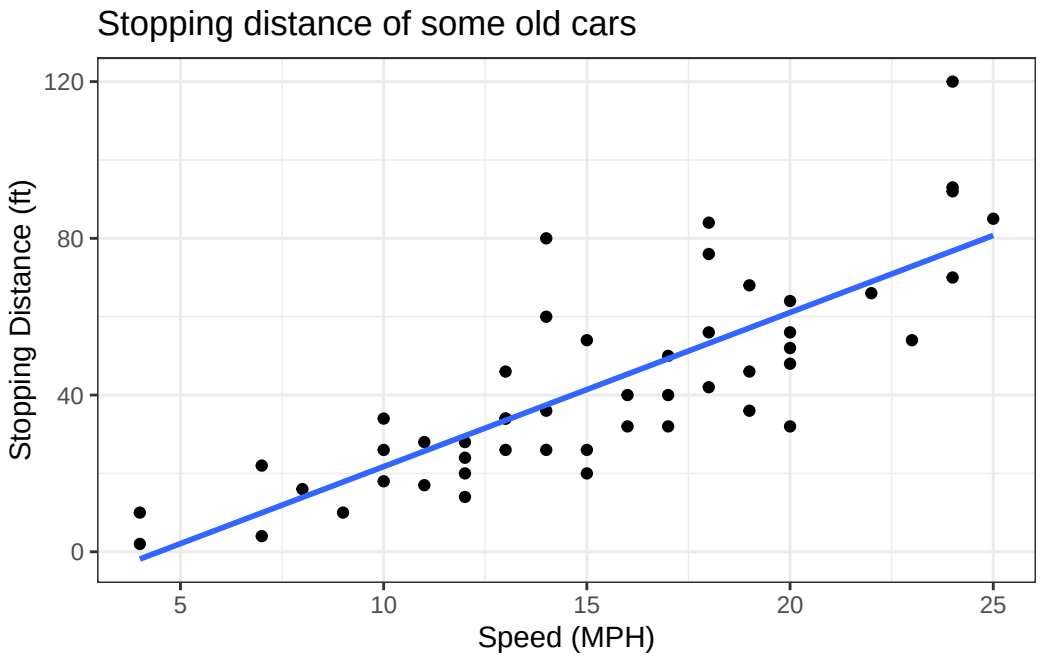
```
ggplot(cars) +
  aes(x=speed, y=dist) +
  geom_point() +
  geom_smooth()
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```




```
ggplot(cars) +
  aes(x=speed, y=dist) +
  geom_point() +
  geom_smooth(method="lm", se=FALSE) +
  theme_bw() +
  labs(x="Speed (MPH)",
       y="Stopping Distance (ft)",
       title="Stopping distance of some old cars")
```

```
`geom_smooth()` using formula = 'y ~ x'
```



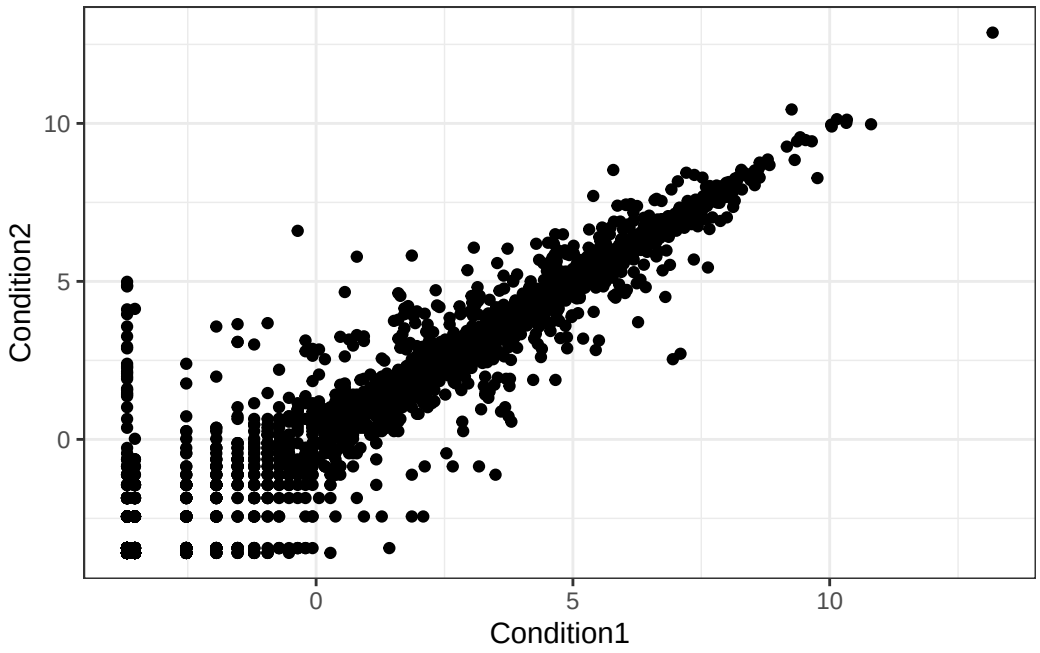
Gene Expression Plot

First we will read some expression data for the effect of a GLP-1 drug

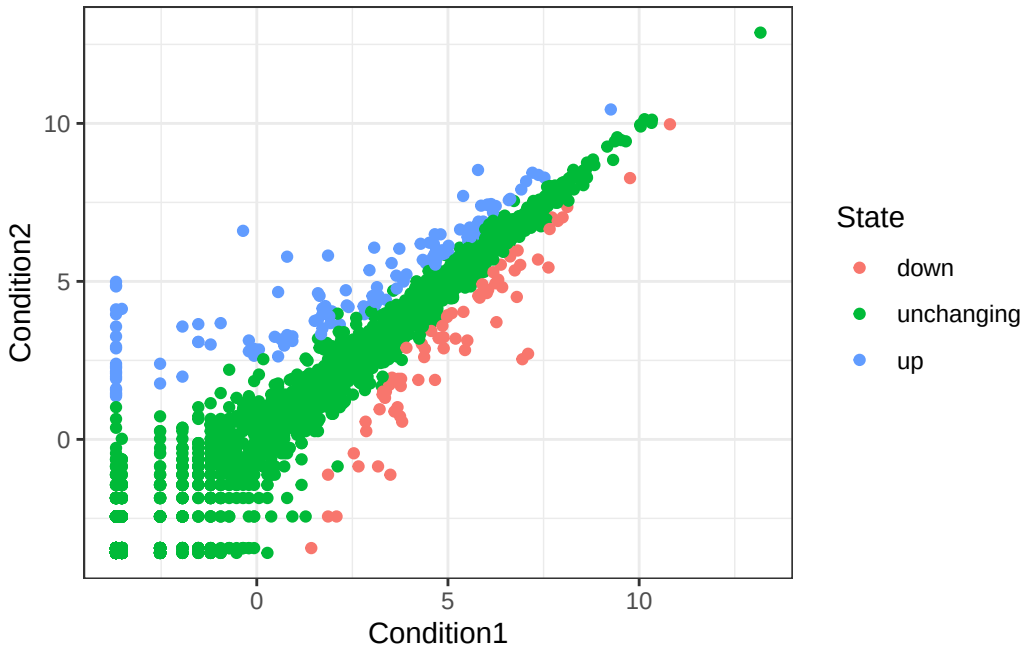
```
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes)
```

```
        Gene Condition1 Condition2      State
1      A4GNT -3.6808610 -3.4401355 unchanging
2       AAAS  4.5479580  4.3864126 unchanging
3      AASDH  3.7190695  3.4787276 unchanging
4       AATF  5.0784720  5.0151916 unchanging
5       AATK  0.4711421  0.5598642 unchanging
6 AB015752.4 -3.6808610 -3.5921390 unchanging
```

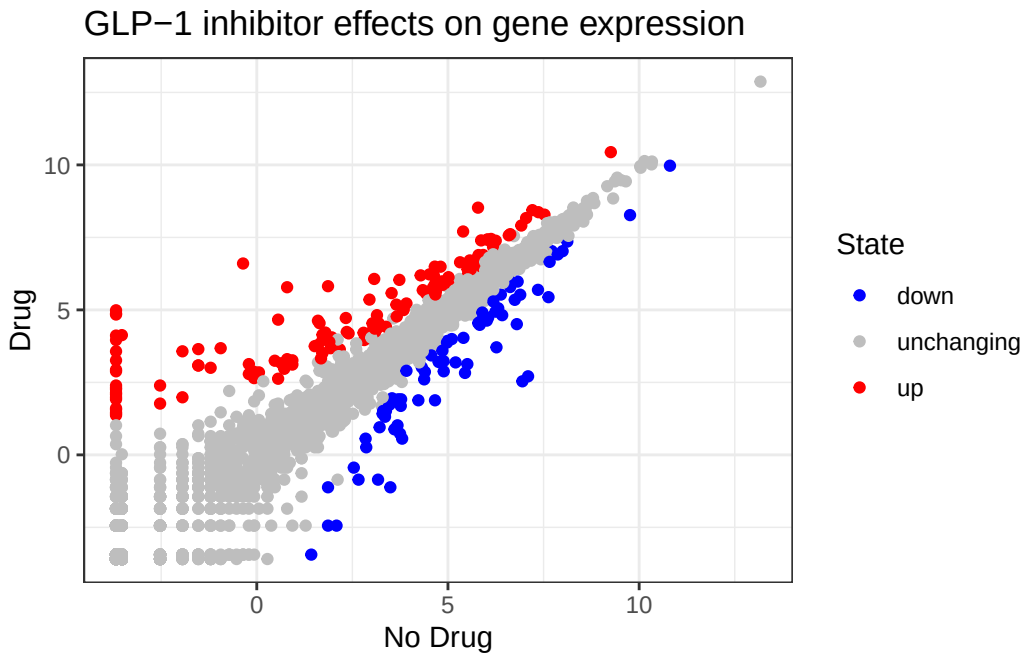
```
ggplot(genes) +
  aes(x=Condition1, y=Condition2) +
  geom_point() +
  theme_bw()
```




```
ggplot(genes) +
  aes(x=Condition1, y=Condition2, col=State) +
  geom_point() +
  theme_bw()
```




```
ggplot(genes) +
  aes(x=Condition1, y=Condition2, col=State) +
  geom_point() +
  theme_bw() +
  scale_colour_manual(values=c("blue", "gray", "red")) +
  labs(x="No Drug", y="Drug", title="GLP-1 inhibitor effects on gene expression")
```



Number of Genes

```
nrow(genes)
```

```
[1] 5196
```

Number of columns

```
ncol(genes)
```

```
[1] 4
```

Column Names

```
colnames(genes)
```

```
[1] "Gene"       "Condition1" "Condition2" "State"
```

```
table(genes$State)
```

```
      down unchanging         up
        72       4997        127
```

```
n.total <- nrow(genes)
n.up <- sum(genes$State == "up")

round(n.up/n.total*100, 2)
```

```
[1] 2.44
```

Going further

Here we read the famous *gapminder* dataset:

```
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.ts

gapminder <- read.delim(url)
head(gapminder)
```

```
      country continent year lifeExp      pop gdpPercap
1 Afghanistan      Asia 1952  28.801  8425333  779.4453
2 Afghanistan      Asia 1957  30.332  9240934  820.8530
3 Afghanistan      Asia 1962  31.997 10267083  853.1007
4 Afghanistan      Asia 1967  34.020 11537966  836.1971
5 Afghanistan      Asia 1972  36.088 13079460  739.9811
6 Afghanistan      Asia 1977  38.438 14880372  786.1134
```

Q. How many entries are in this dataset?

```
nrow(gapminder)
```

```
[1] 1704
```

Q. How many different "country" values are in this dataset?

```
unique(gapminder$country)
```

```
 [1] "Afghanistan"              "Albania"
 [3] "Algeria"                  "Angola"
 [5] "Argentina"                "Australia"
 [7] "Austria"                  "Bahrain"
 [9] "Bangladesh"               "Belgium"
[11] "Benin"                    "Bolivia"
[13] "Bosnia and Herzegovina"   "Botswana"
[15] "Brazil"                   "Bulgaria"
[17] "Burkina Faso"             "Burundi"
[19] "Cambodia"                 "Cameroon"
[21] "Canada"                   "Central African Republic"
[23] "Chad"                     "Chile"
[25] "China"                    "Colombia"
[27] "Comoros"                  "Congo, Dem. Rep."
[29] "Congo, Rep."              "Costa Rica"
[31] "Cote d'Ivoire"            "Croatia"
[33] "Cuba"                     "Czech Republic"
[35] "Denmark"                  "Djibouti"
[37] "Dominican Republic"       "Ecuador"
[39] "Egypt"                    "El Salvador"
[41] "Equatorial Guinea"        "Eritrea"
[43] "Ethiopia"                 "Finland"
[45] "France"                   "Gabon"
[47] "Gambia"                   "Germany"
[49] "Ghana"                    "Greece"
[51] "Guatemala"                "Guinea"
[53] "Guinea-Bissau"            "Haiti"
[55] "Honduras"                 "Hong Kong, China"
[57] "Hungary"                  "Iceland"
[59] "India"                    "Indonesia"
[61] "Iran"                     "Iraq"
[63] "Ireland"                  "Israel"
[65] "Italy"                    "Jamaica"
[67] "Japan"                    "Jordan"
[69] "Kenya"                    "Korea, Dem. Rep."
[71] "Korea, Rep."              "Kuwait"
[73] "Lebanon"                  "Lesotho"
[75] "Liberia"                  "Libya"
[77] "Madagascar"               "Malawi"
[79] "Malaysia"                 "Mali"
[81] "Mauritania"               "Mauritius"
[83] "Mexico"                   "Mongolia"
[85] "Montenegro"               "Morocco"
```

```
 [87] "Mozambique"               "Myanmar"
 [89] "Namibia"                  "Nepal"
 [91] "Netherlands"              "New Zealand"
 [93] "Nicaragua"                "Niger"
 [95] "Nigeria"                  "Norway"
 [97] "Oman"                     "Pakistan"
 [99] "Panama"                   "Paraguay"
[101] "Peru"                     "Philippines"
[103] "Poland"                   "Portugal"
[105] "Puerto Rico"              "Reunion"
[107] "Romania"                  "Rwanda"
[109] "Sao Tome and Principe"    "Saudi Arabia"
[111] "Senegal"                  "Serbia"
[113] "Sierra Leone"             "Singapore"
[115] "Slovak Republic"          "Slovenia"
[117] "Somalia"                  "South Africa"
[119] "Spain"                    "Sri Lanka"
[121] "Sudan"                    "Swaziland"
[123] "Sweden"                   "Switzerland"
[125] "Syria"                    "Taiwan"
[127] "Tanzania"                 "Thailand"
[129] "Togo"                     "Trinidad and Tobago"
[131] "Tunisia"                  "Turkey"
[133] "Uganda"                   "United Kingdom"
[135] "United States"            "Uruguay"
[137] "Venezuela"                "Vietnam"
[139] "West Bank and Gaza"       "Yemen, Rep."
[141] "Zambia"                   "Zimbabwe"
```

```
length(unique(gapminder$country))
```

```
[1] 142
```

```
table(gapminder$country)
```

```
     Afghanistan              Albania              Algeria
              12                   12                   12
          Angola            Argentina            Australia
              12                   12                   12
         Austria              Bahrain           Bangladesh
```

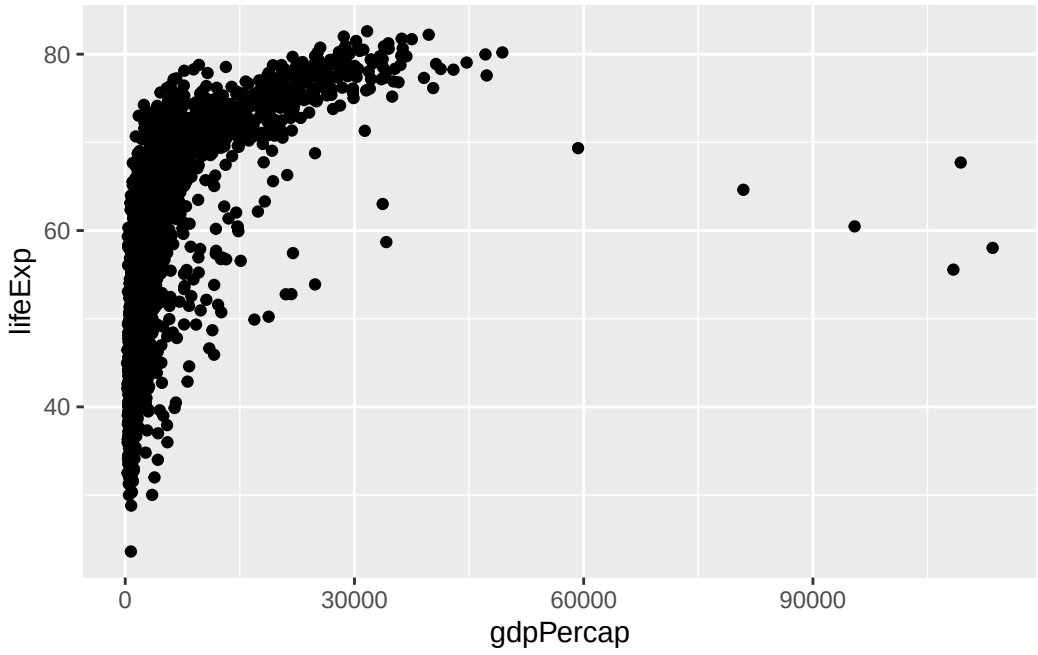
12	12	12
Belgium	Benin	Bolivia
12	12	12
Bosnia and Herzegovina	Botswana	Brazil
12	12	12
Bulgaria	Burkina Faso	Burundi
12	12	12
Cambodia	Cameroon	Canada
12	12	12
Central African Republic	Chad	Chile
12	12	12
China	Colombia	Comoros
12	12	12
Congo, Dem. Rep.	Congo, Rep.	Costa Rica
12	12	12
Cote d'Ivoire	Croatia	Cuba
12	12	12
Czech Republic	Denmark	Djibouti
12	12	12
Dominican Republic	Ecuador	Egypt
12	12	12
El Salvador	Equatorial Guinea	Eritrea
12	12	12
Ethiopia	Finland	France
12	12	12
Gabon	Gambia	Germany
12	12	12
Ghana	Greece	Guatemala
12	12	12
Guinea	Guinea-Bissau	Haiti
12	12	12
Honduras	Hong Kong, China	Hungary
12	12	12
Iceland	India	Indonesia
12	12	12
Iran	Iraq	Ireland
12	12	12
Israel	Italy	Jamaica
12	12	12
Japan	Jordan	Kenya
12	12	12
Korea, Dem. Rep.	Korea, Rep.	Kuwait
12	12	12

Lebanon	Lesotho	Liberia
12	12	12
Libya	Madagascar	Malawi
12	12	12
Malaysia	Mali	Mauritania
12	12	12
Mauritius	Mexico	Mongolia
12	12	12
Montenegro	Morocco	Mozambique
12	12	12
Myanmar	Namibia	Nepal
12	12	12
Netherlands	New Zealand	Nicaragua
12	12	12
Niger	Nigeria	Norway
12	12	12
Oman	Pakistan	Panama
12	12	12
Paraguay	Peru	Philippines
12	12	12
Poland	Portugal	Puerto Rico
12	12	12
Reunion	Romania	Rwanda
12	12	12
Sao Tome and Principe	Saudi Arabia	Senegal
12	12	12
Serbia	Sierra Leone	Singapore
12	12	12
Slovak Republic	Slovenia	Somalia
12	12	12
South Africa	Spain	Sri Lanka
12	12	12
Sudan	Swaziland	Sweden
12	12	12
Switzerland	Syria	Taiwan
12	12	12
Tanzania	Thailand	Togo
12	12	12
Trinidad and Tobago	Tunisia	Turkey
12	12	12
Uganda	United Kingdom	United States
12	12	12
Uruguay	Venezuela	Vietnam

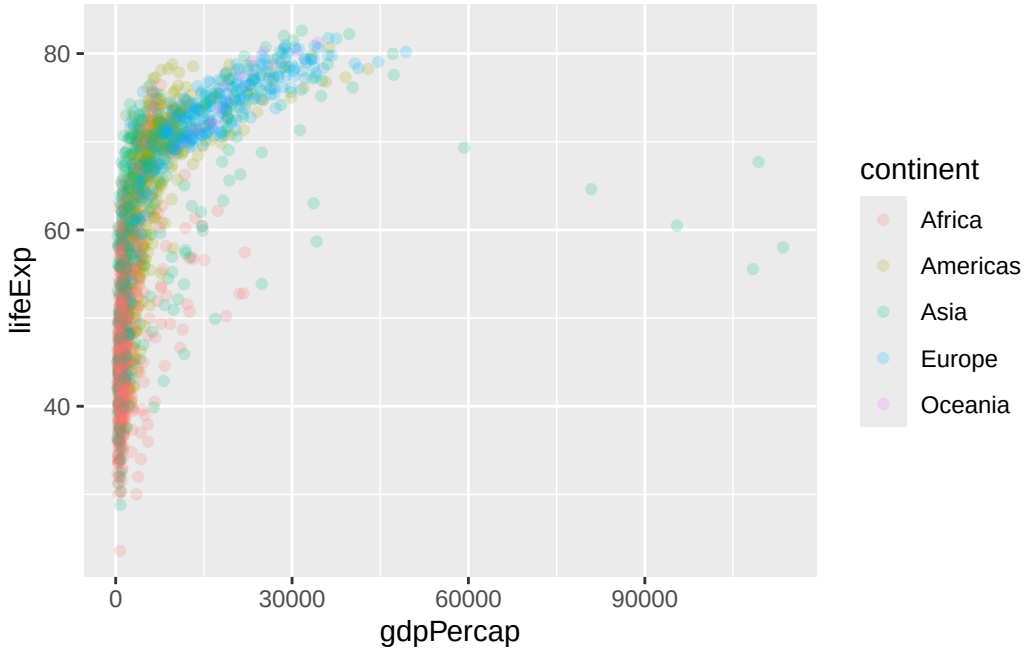
```
                  12                          12                          12
West Bank and Gaza                Yemen, Rep.                      Zambia
                  12                          12                          12
            Zimbabwe
                  12
```

Let's make our first plot of `gdpPercap` vs `lifeExp`:

```
ggplot(gapminder) +
  aes(x=gdpPercap, y=lifeExp) +
  geom_point()
```

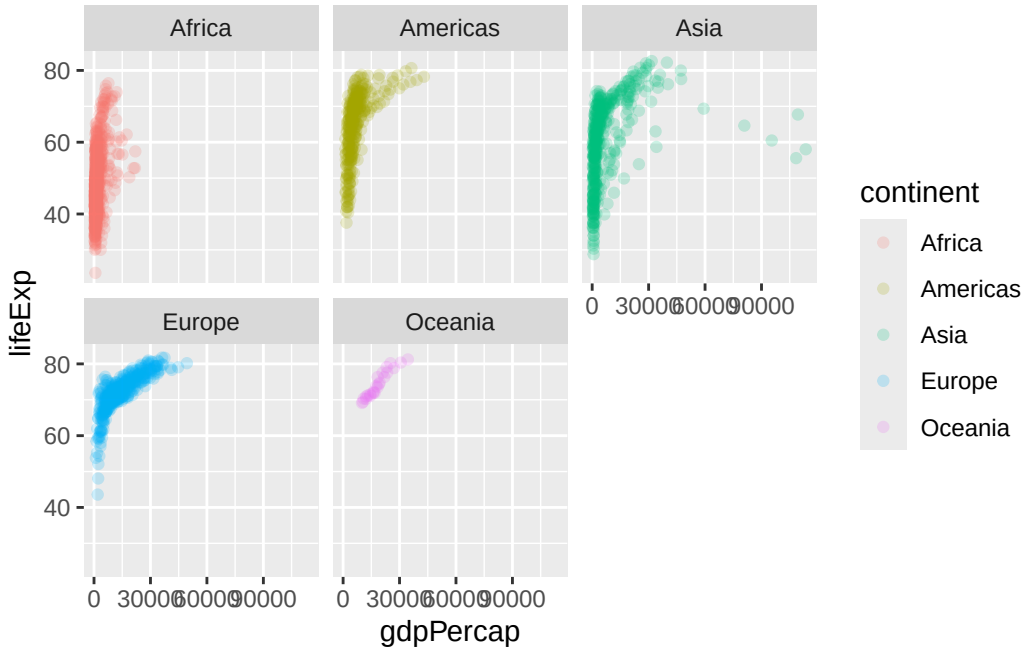



```
ggplot(gapminder) +
  aes(x=gdpPercap, y=lifeExp, col=continent) +
  geom_point(alpha=0.2)
```



One useful layer we can add to explore big datasets like this is called "facetting"

```
ggplot(gapminder) +
  aes(x=gdpPercap, y=lifeExp, col=continent) +
  geom_point(alpha=0.2) +
  facet_wrap(~continent)
```



Limit to 1977 and 2007

We can install and use the **dplyr** package to help "filter" the full dataset down to a smaller subset that only includes the years we want.

First, we need to run `install.packages("dplyr")` and load it with `library(dplyr)`.

```
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
filter(gapminder, year==2007, country=="Canada")
```

```
  country continent year lifeExp      pop gdpPercap
1  Canada  Americas 2007  80.653 33390141  36319.24
```

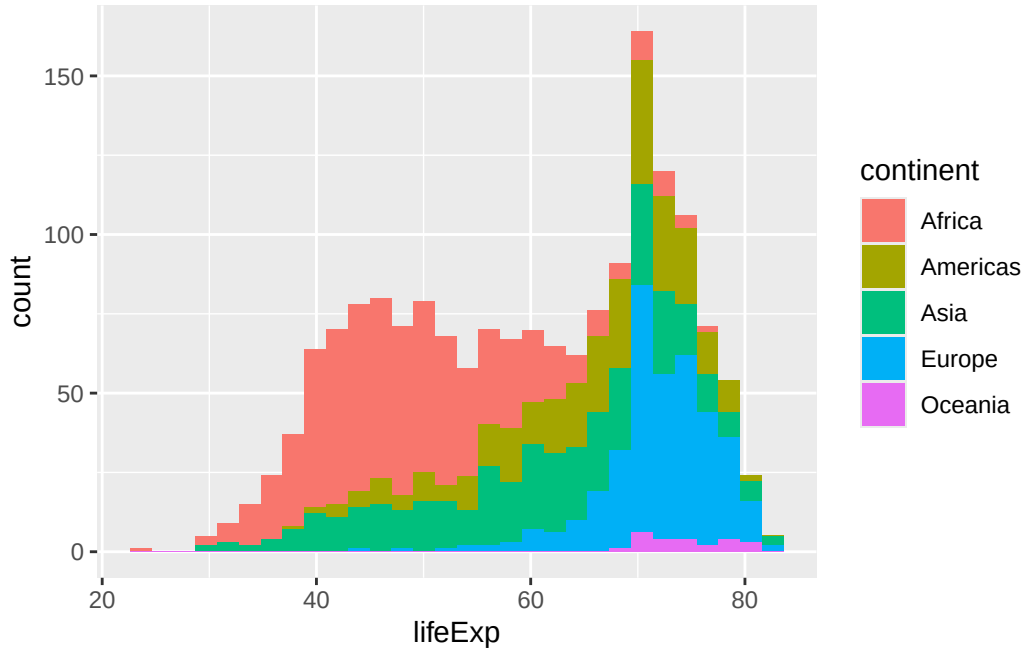
```
filter(gapminder, year==1977, country=="Canada")
```

```
  country continent year lifeExp      pop gdpPercap
1  Canada  Americas 1977   74.21 23796400  22090.88
```

Q. Make a histogram of lifeExp

```
ggplot(gapminder) +
  aes(lifeExp, fill=continent) +
  geom_histogram()
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



Boxplot

```
ggplot(gapminder) +
  aes(lifeExp, fill=continent) +
  geom_boxplot()
```

